INCH-POUND

MIL-S-13949/27A
18 May 1993
SUPERSEDING
MIL-S-13949/27
4 October 1991

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL SC
(WOVEN S-2 GLASS FIBER 1/ REINFORCEMENT, MAJORITY CYANATE ESTER RESIN,
FLAME RESISTANT, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: S for woven electrical grade S-2 glass fiber. 1/ This reinforcement material is a continuous filament yarn with a chemical composition by weight that is within the following limits:

Al_2O_3	24 - 26 percent
SiO_2	64 - 66 percent
MgO	9 - 11 percent
Na_2O and K_2O	0 - 0.3 percent
Fe_2O_3	0 - 0.3 percent

Composition to be certified by the yarn supplier as requested. All other parameters shall be as detailed in table I.

Weave style:

Plain weave: A fabric configuration where each warp end should go over one pick and under the next, and each pick should go over one warp end and under the next.

Satin weave: A fabric configuration where the surface is almost made up entirely of warp or filling floats over all but one of a designated number of adjacent yarns, thereby producing a smooth surface. The intersection points do not fall in a straight diagonal (or twill), but in pattered formation. For example, in a four harness satin weave pattern, there is a three by one interlacing, whereby, a filling yarn floats over three warp yarns and under one.

Resin system: C in accordance with MIL-S-13949.

1/ The term "S-2 glass fiber" is a registered trademark of the Owens-Corning Fiberglass Corporation. The term "S-2 glass fiber" is used herein in a generic sense and not in the sense of a registered trademark to identify a type of reinforcement material used in the production of printed wiring board materials. The use of the term "S-2 glass fiber" herein does not impose a requirement to utilize Owens-Corning's "S-2 glass fiber" material in the production of printed wiring board materials manufactured in accordance with this specification sheet, although the use of the Owens-Corning material would meet this specification sheet's requirement for an "S-2 glass fiber" reinforcement material.

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TABLE I. Woven reinforcement construction characteristics.

Style	Thickness (inches)	Thread count	Weight, maximum (±10 percent) (ounces per square yard)
6080	.0021	60 x 47	1.38
6112	.0036	40 x 39	2.08
6116	.0040	60 x 58	3.09
6120 1/	.0038	60 x 58	3.09
6180 1/	.0020	60 x 60	1.55
6313	.0035	60 x 64	2.38
6314 1/	.0033	60 x 64	2.38
6413	.0034	60 x 56	2.30
6628	.0068	44 x 32	6.03

1/ These styles are a four harness satin weave. All other styles are plain weave.

Special consideration (third letter designator): N in accordance with MIL-S-13949.

Type of metal foil: Any as specified in accordance with MIL-S-13949.

Weight of metal foil: H, 1, 2, or O in accordance with MIL-S-13949.

Solderability: Base materials clad with metal foil types A, B, C, D, G, H, J, K, L, and M shall be tested in accordance with MIL-S-13949. Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist: C or X in accordance with MIL-S-13949.

Grade of pits or dents: In accordance with MIL-S-13949.

Qualification inspection routine. The inspection routine shall include the glass transition temperature test inaccordance with MIL-S-13949.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values:

Dimensional stability (for base material thicknesses less than .020 inch (0.51 mm)).

After bake: Class A, .00075 inch/inch (0.00075 mm/mm) maximum; class B, .00050 inch/inch (0.00050 mm/mm) maximum.

After stress: Class A, .00075 inch/inch (0.00075 mm/mm) maximum; class B, .00050 inch/inch (0.00050 mm/mm) maximum.

Flammability.

Duration of burning time, each flame application:

Base material thicknesses greater than or equal to .020 inch (0.51 mm).

Average: 25 seconds maximum.

Single: 30 seconds maximum.

Base material thicknesses less than .020 inch (0.51 mm): 15 seconds maximum.

Burn length, average: 12 inches (305 mm) maximum.

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Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm)).

Crosswise: 50,000 pounds per square inch minimum.

Lengthwise: 50,000 pounds per square inch minimum.

Lengthwise at elevated temperature of 204°C: 40,000 pounds per square inch minimum.

Glass transition temperature: 230°C minimum.

Peel strength values for copper foil: See table II.

TABLE II. Peel strength values, minimum pounds per inch of width.

Test and copper foil weight	Base material thickness	
	Less than .020 inch	.020 inch and over
After thermal stress:		
½-oz/ft² (designator H)	4.0	4.0
1-oz/ft² (designator 1)	4.5	4.5
2-oz/ft² (designator 2)	5.5	5.5
At elevated temperature of 170°C:		
½-oz/ft² (designator H)	4.0	4.0
1-oz/ft² (designator 1)	4.5	4.5
2-oz/ft² (designator 2)	5.5	5.5
After exposure to processing solutions:		
½-oz/ft² (designator H)	4.0	4.0
1-oz/ft² (designator 1)	4.5	4.5
2-oz/ft² (designator 2)	5.5	5.5

Arc resistance: 120 seconds minimum.

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 40 kilovolts minimum.

Electric strength, average (for base material thicknesses less than .020 inch (0.51 mm)): 750 volts/mil minimum.

Loss tangent at 1 MHz: .015 maximum.

Permittivity at 1 MHz: 4.3 maximum.

Surface resistivity:

After moisture resistance conditioning: 10^4 megohm minimum.

After elevated temperature (125°C) conditioning: 10^3 megohm minimum.

Volume resistivity:

After moisture resistance conditioning: 10^6 megohm-cm minimum.

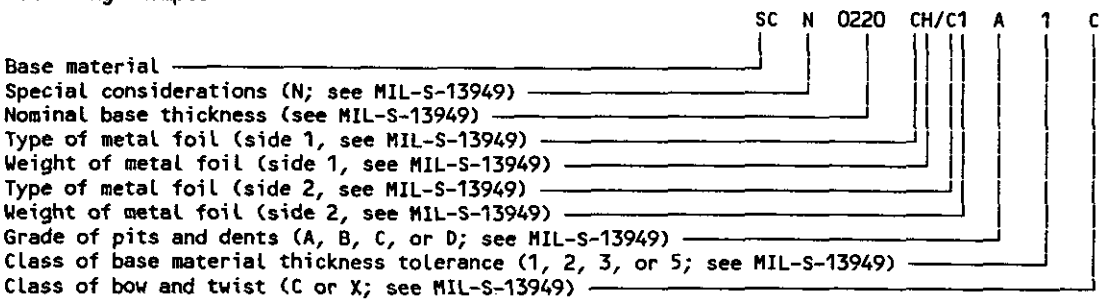
After elevated temperature (125°C) conditioning: 10^3 megohm-cm minimum.

Water absorption (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 1.0 percent maximum.

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Supersession information: Not applicable.

Type designation: Consists of "SC" prefix followed by coded alphas and numeric characters as shown in the following example:



Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
NSA - NS
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0032-01)

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